

CYBER SECURITY

UNIT 2 – COMPLETE DETAILED NOTES

Web Jacking • Online Frauds • Software Piracy • Computer Network Intrusions •
Password Sniffing • Identity Theft

Cyber Terrorism • Virtual Crime • Perception of Cyber Criminals • Web Server Hacking • Session Hijacking

Book-style notes with Practical Examples, Definitions, Flowcharts,
Diagrams, Important Points, Exam Notes & Memory Tricks

UNIT 2 – SYLLABUS (As per Examination)

S.No	Unit 2 Topics	Marks
1	Web Jacking	---
2	Online Frauds	---
3	Software Piracy	---
4	Computer Network Intrusions	---
5	Password Sniffing	---
6	Identity Theft	---
7	Cyber Terrorism	---
8	Virtual Crime	---
9	Perception of Cyber Criminals (Hackers, Insurgents & Extremist Groups)	---
10	Web Server Hacking	---
11	Session Hijacking	---

How to read this book: Har topic me sabse pehle ek **long practical example** diya hai, phir **short definition**, phir **flowchart / diagram**, phir **important points** aur aakhir me **exam notes**. Isi order me padhna — definition yaad aasan ho jayegi.

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PART 1 – WEB JACKING

Unit 2 ka pehla topic. Web Jacking me website ki **poori control** chali jati hai — website khud nahi badalti, par uske visitors ko kahin aur bhej diya jata hai. Detail me samjhte hain.

1.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek chhoti company **Bandhan Store** ki website thi — bandhanstore.com — jahan se log clothes kharidte the aur card se payment karte the.

Ek din ek hacker ne company ke **domain/DNS account ka password** hack kar liya. Ab usne:

1. Website ka saara control apne paas le liya.
2. Visitors jab bhi bandhanstore.com kholte the, wo seedha **fake page** par chale jaate the jo bilkul asli jaisa dikhta tha.
3. Fake page par customer ne card details daali — details hacker ke paas chali gayi.
4. Kuch visitors ko virus wali file download bhi ho gayi.

Company ko pata bhi nahi chala ki uski website ab hacker ke control me hai. **Website ko hi 'jacked' (control me le liya) kar ke visitors ko fake jagah bhejna — yehi Web Jacking hai.**

1.2 Definition

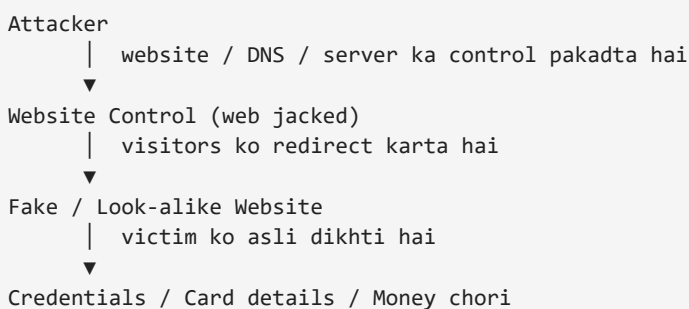
DEFINITION

Web Jacking ek attack hai jisme attacker kisi website ka **control haath me le leta hai** aur uske saare visitors ko bina unki jaankari ke kisi **fake ya illegal website par redirect** kar deta hai — taaki data, paise ya credentials chura sake.

Simple: Website ki gaddi ka steering wheel chura kar uske sawari ko kahin aur le jaana.

1.3 Flowchart – Web Jacking Kaise Hota Hai

FLOWCHART



1.4 Web Jacking Ke Tarike (Kaise Hota Hai)

- **DNS Hijacking** — domain/DNS settings badal kar website ka traffic kahin aur bhejna.
- **Web server compromise** — server hack karke redirect rules (htaccess/config) change karna.
- **Malicious redirects** — site me code/malware daal kar visitors ko fake page par le jaana.
- **Look-alike domain + traffic steal** — milta-julta domain (bandhanstore.net) banakar confusion paida karna.

1.5 Web Jacking vs Website Defacement

Basis	Web Jacking	Defacement
Kya hota hai	Website ka control + visitors redirect	Sirf page ka content badal dena

Visitors	Fake site par le jaaye jaate hain	Asli site par rehte hain
Data theft	Credentials/card details chori hoti hai	Aam taur par sirf message/chinta
Damage	Financial loss, identity theft	Reputation + trust loss

1.6 Prevention (Bachao Ke Tarike)

- Domain registrar account par **strong password + 2FA**.
- **DNSSEC** enable karo (DNS ko fake responses se bachata hai).
- Website par **HTTPS** lagao aur hosting passwords strong rakho.
- Server/software ko **update** karte raho, backups rakho.
- Website aur DNS records ko **regularly monitor** karo.
- Visitors ko bhi sikhana: URL dhyan se dekho, koi bhi link par OTP/card daalne se pehle check karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Web Jacking = website ka control le kar visitors ko fake site par redirect karna.
2. Main tarika = DNS hijacking ya web server compromise.
3. Defacement me sirf content badalta hai, Web Jacking me poora control + redirect.

Viva Q1. Web jacking me visitor ko kya dikhta hai?

Ans: Fake/look-alike page jo asli jaisa dikhta hai.

Viva Q2. Web jacking se kaise bachein?

Ans: DNSSEC, strong password + 2FA, HTTPS, monitoring.

MEMORY TRICK

Web Jacking = Website ki hijack. Jaise plane hijack kar ke usse kahin aur le jaate hain, waise hi website hijack kar ke uske visitors ko fake page par le jate hain.

PART 2 – ONLINE FRAUDS

Online Fraud sabse aam cyber crime hai — **paisa lootne ke liye dhoka**. Is part me hum types aur bachao ka tarika detail me dekhenge.

2.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Priya ko WhatsApp par message aaya: “Amazon se aapki ek order wapas ho gayi hai. Rs. 2,499 ka refund milega. Form bharein.”

Usne form bhara — usme **card number, expiry, CVV, OTP** maanga gaya. Dalte hi uske card se **Rs. 15,000** kat gaye.

Kuch din baad usne ek website se **Rs. 999 ka smartphone** dekha (asli price Rs. 9,000). Payment online kiya — phone kabhi nahi aaya, website gayab. Ye dono **Online Frauds** hain.

Ek aur example: Ravi ko job ka email mila — “Registration fee Rs. 5,000 bharo, job pakki.” Fee bhari, interview ki jagah form bharnе ko bola, phir baat hi khatam. **Ye bhi Online Fraud hai.**

2.2 Definition

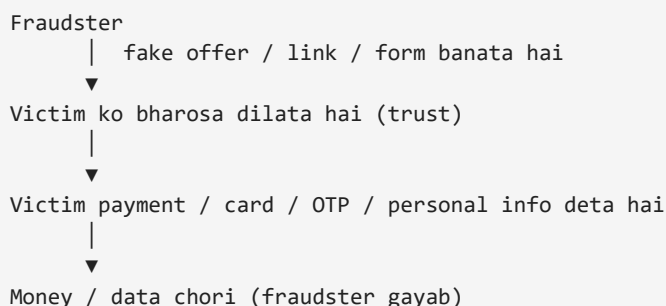
DEFINITION

Online Fraud woh dhoka (fraud) hai jo **internet ke through** kiya jata hai — jisme victim ko jhootha bharosa de kar usse **paise, goods, ya personal information** illegally haasil ki jati hai.

Simple: Internet par bana hua jhootha bharosa — paisa lootne ki koshish.

2.3 Flowchart – Online Fraud Kaise Hota Hai

FLOWCHART



2.4 Types of Online Frauds

Type	Kya hota hai
Phishing	Fake email/site se credentials churana
Lottery / Prize scam	Jhootha inaam dikha kar fee vasoolna
UPI / UPI-refund fraud	Fake UPI link ya refund ke bahane OTP lena
Card fraud	Card details + OTP chura kar kharidaari
Job scam	Job ke naam par registration fee lena
Auction / Shopping scam	Fake site par payment lena, samaan nahi dena
Romance scam	Pyara wala chat karke paise maangna
Donation scam	Fake charity ke naam par daan maangna

2.5 Red Flags (Fraud Kaise Pehchano)

- **Too good to be true** — itna sasta / itna bada inaam aam nahi milta.
- **Jaldi bharo** wali pressure (time limit).
- **OTP / CVV / PIN** maangna — asli company kabhi nahi maangti.
- Payment sirf **personal account** me bhejna.
- **Fake URL / site** — spelling check karo (amaz0n.com).
- **Unverified phone/email** — Google par search karke verify karo.

2.6 Prevention (Bachao Ke Tarike)

- OTP/PIN/CVV **kabhi kisi ko na do** — bank/company bhi nahi.
- Sirf **trusted sites** se online payment karo.
- Website ka **URL + padlock** (HTTPS) check karo.
- **Unexpected links** par click na karo.
- Shopping ke liye **verified sellers** hi chuno.
- Fraud hone par turant **bank + cyber cell (1930)** ko report karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Online Fraud = internet ke through kiya gaya dhoka, motive = paisa/info.
2. Sabse aam = phishing, UPI fraud, lottery scam, job scam.
3. Golden rule: **OTP/CVV/PIN kabhi share mat karo.**

Viva Q1. Online fraud me victim ko kya dilaya jata hai?

Ans: Jhootha bharosa (trust).

Viva Q2. Fraud hone par kya karein?

Ans: Turant bank ko inform + 1930 par report + evidence save karna.

MEMORY TRICK

Online Fraud = Jhootha wala dosti. Jaise koi dost ban ke baad loot le — waise hi scammer bharosa jeet kar paisa lootta hai. Rule: **kisi ko bhi OTP mat do.**

PART 3 – SOFTWARE PIRACY

Software Piracy me software ka **illegal copy** banakar usse bechna / use karna hota hai. Ye developer company ka **legal right (copyright)** todta hai.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Aman ek computer shop chalata hai. Wo har naye laptop par **Windows + Photoshop** install karta hai — bina license kharede. Customer ko laptop mil jata hai, shop ko sale mil jaati hai, lekin Microsoft aur Adobe ko **unka paisa nahi milta**.

Ek baar Aman ne isi software ko **WhatsApp group me bhej diya** taki 20 customers khud install kar lein. Ek customer ka **crack wala file download** karte hi uske PC par virus aaya aur uske bank details chori ho gayi.

Software company ne Aman ke khilaf **legal case** kiya — jaise kisi aur ki cheez churana illegal hai, waise software ka copy banana bhi illegal hai. **Yehi Software Piracy hai**.

3.2 Definition

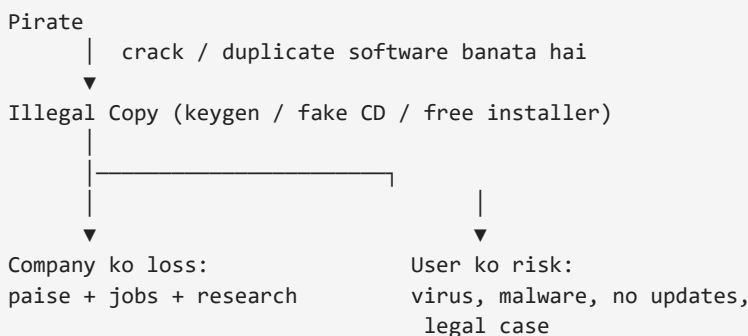
DEFINITION

Software Piracy software ka **unauthorized copying, distribution, sale ya use** hai — bina uske owner (developer company) ki proper license / permission ke.

Simple: Jo software paise de kar kharidna chahiye, usse free me copy/bech kar company ka paisa kamaana / bachana.

3.3 Flowchart

FLOWCHART



3.4 Types of Software Piracy

Type	Kya hota hai
Softlifting	1 license ko bahut saare computers par install karna
Client-server overuse	Network me allowed se zyada users server par use karna
Hard-disk loading	Shop naye PC par bina license software install kar deti hai
Counterfeiting	Nakli CD/DVD (fake look-alike) bechna
Online piracy	Crack / keygen / free download sites se distribution

3.5 Piracy Ka Nuksan (Impact)

- **Developer company** ko paise nahi milte — research aur jobs khatam.

- **User ke liye khatra** — cracked files me virus/malware chhupa hota hai.
- **No updates / support** — software risky rehta hai.
- **Legal action** — fine aur jail ho sakta hai.
- Desh ki **economy** ko bhi loss.

3.6 Prevention (Bachao Ke Tarike)

- Sirf **licensed software** kharido (official website / authorised shop).
- **Product key / activation** ke systems ka use karo.
- **DRM (Digital Rights Management)** — software ko copy hone se rokna.
- Company me **software audit** — kaun sa licensed, kaun sa nahi.
- Public awareness — piracy bhi chori hai.
- Open-source / free alternatives (Linux, LibreOffice) use karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Software Piracy = software ka bina license copy / bech / use karna.
2. Types = softlifting, hard-disk loading, counterfeiting, online piracy, client-server overuse.
3. Risk = company ka paisa loss + user ke PC par malware.

Viva Q1. Softlifting kya hai?

Ans: Ek license ko kai computers par install karna.

Viva Q2. Piracy se user ko kya risk hai?

Ans: Cracked file me virus/malware, no updates, legal case.

MEMORY TRICK

Software Piracy = Book ki photo-copy. Jaise book ka photocopy banakar bechna illegal hai, waise software ka copy banakar bechna/use karna illegal hai.

PART 4 – COMPUTER NETWORK INTRUSIONS

Network Intrusion matlab **bina permission ke kisi network ke andar ghusna**. Jaise ghar me andar ghus kar samaan dekhna, waise network me ghus kar data par nazar aur haath.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek company ke office me 100 computers ek network se jude hain. Ek din ek hacker ne **network scan** kiya aur paaya ki ek **purana server** khula (open port) hai jiska password **admin123** tha.

Hacker ne us password se login kiya — ab wo **network ke andar** tha. Usne:

1. Salary records chura liye.
2. Ek system par **backdoor** install kar diya.
3. Aur do computers se bhi paise-transfer files nikali.

Company ko pata hi nahi chala. Jab tak software update hua, data nikal chuka tha. **Bina permission ke network ke andar ghus kar data churana / malware daalna — Computer Network Intrusion.**

4.2 Definition

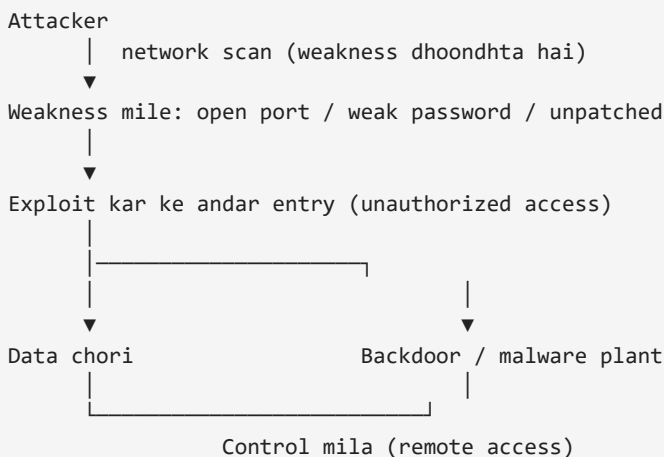
DEFINITION

Computer Network Intrusion kisi computer network ya system me **unauthorized access** hai — jisme attacker security tod kar andar ghus kar data churata hai, malware daalta hai, ya network ko apne control me le leta hai.

Simple: Network ki deewar (security) tod kar andar ghusna.

4.3 Flowchart – Intrusion Process

FLOWCHART



4.4 Types of Network Intrusions

Type	Kya hota hai
External Intrusion	Bahar se hacker network me ghusna
Internal Intrusion	Andar ka employee/insider misuse karna
Host Intrusion	Kisi ek computer/system par unauthorized access
Network Intrusion	Poore network me unauthorized entry

4.5 Intrusion Kaise Hoti Hai (Methods)

- **Weak passwords** — admin123, password, 123456.
- **Unpatched / old software** — known vulnerabilities khuli hain.
- **Open ports** — services internet par khuli hain.
- **Phishing / social engineering** — employee se login details nikalna.
- **Brute force** — password guess karna.
- **Malware** — keylogger/trojan se andar ka rasta.

4.6 Detection aur Prevention

- **IDS / IPS** (Intrusion Detection / Prevention System) lagaana.
- **Firewall** — bahar ka traffic filter karna.
- Software ko **patch / update** karte rehna.
- **Strong passwords + MFA.**
- **Network segmentation** — sensitive part alag rakhna.
- **Logs monitoring** — suspicious activity track karna.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Network Intrusion = bina permission ke network/system me unauthorized access.
2. Sabse aam tarika = weak password / open port / unpatched software.
3. Detect karte hain = IDS/IPS, firewall, log monitoring.

Viva Q1. Intrusion me attacker entry ke baad kya karta hai?

Ans: Data chori, malware/backdoor plant, remote control.

Viva Q2. Intrusion se bachne ka best tarika?

Ans: Firewall + IDS/IPS + patch updates + strong password.

MEMORY TRICK

Network Intrusion = Chor ka ghar me ghusna. Jaise chor diwaar tod kar ghus kar samaan le jata hai, waise hacker security tod kar data le jata hai.

PART 5 – PASSWORD SNIFFING

Password Sniffing me attacker **chupke se network traffic sunta hai** aur login password pakad leta hai — victim ko kuch pata nahi chalta.

5.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Rahul ek **cafe ki free WiFi** par baitha tha. Usi cafe me ek hacker bhi tha jisne apna laptop laga rakha tha jisme **packet sniffing tool (Wireshark)** chal raha tha.

Rahul ne apne account par login kiya — lekin wo site **HTTP** par chal rahi thi (secure nahi).

Hacker ke tool ne Rahul ke **username aur password** ko network ke packets me padh liya — kyunki data bina encryption ke travel kar raha tha.

Agle hi din hacker ne Rahul ke account se login kiya aur **Rs. 8,000** ka recharge/fraud kar diya.

Chupke se network traffic se password pakadna — Password Sniffing.

5.2 Definition

DEFINITION

Password Sniffing network traffic ko monitor karke **passwords aur login credentials ko pakadna** hai — bina victim ko pata chale. Ye aam taur par **unencrypted (plain text) data** par hota hai.

Simple: Victim ka password, hacker ke kaano me gir jata hai — jaise koi baat-chit sun li.

5.3 Flowchart – Sniffing Process

FLOWCHART

```
graph TD
    A[Attacker same network par (WiFi / LAN)] --> B[sniffing tool (Wireshark, tcpdump) chalata hai]
    B --> C[Victim login karta hai (HTTP / unencrypted)]
    C --> D[password plain text me travel karta hai]
    D --> E[Attacker packets ko padh leta hai]
    E --> F[Username + Password captured]
    F --> G[Victim ke account me login / misuse]
```

5.4 Ye Kyu Possible Hai?

IMPORTANT POINTS

Jab bhi data **encrypt nahi hota** (jaise HTTP, Telnet, FTP, old email), to wo network par **plain text** me travel karta hai. Sniffer us traffic ko aasaan se padh leta hai. **HTTPS/SSH encryption hone par data padha nahi ja sakta** — isliye encrypted connection zaroori hai.

5.5 Sniffing Tools (Exam Me Naam Aate Hain)

Wireshark, tcpdump, Cain & Abel, Ettercap — ye network packets pakad kar dikhate hain.

5.6 Prevention (Bachao Ke Tarike)

- **HTTPS / SSL** wali sites hi use karo (URL me padlock).
- **VPN** use karo — traffic encrypt ho jata hai.

- **Public WiFi** par sensitive kaam (banking) na karo.
- WiFi par **WPA2/WPA3** encryption lagao.
- **2FA** lagao — password milne par bhi extra step.
- Passwords har jagah **same mat rakho**.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Password Sniffing = network traffic se passwords pakadna (bina permission).
2. Ye tab possible hai jab data **unencrypted (plain text)** ho.
3. Bachao = HTTPS, VPN, WPA2/WPA3, public WiFi avoid.

Viva Q1. Sniffing me attacker ko kya chahiye?

Ans: Victim ke same network par hona + unencrypted traffic.

Viva Q2. Sniffing se bachne ka best tarika?

Ans: Encrypted connection (HTTPS/SSH/VPN) use karna.

MEMORY TRICK

Sniffing = Kaan lagana. Jaise koi aapke paas aakar baat sun leta hai, waise sniffer network par aakar password sun leta hai. Isliye **baat (data) encrypt karo**.

PART 6 – IDENTITY THEFT

Identity Theft me **aapki pehchan (identity)** chori ho jaati hai — aapka naam, Aadhaar, PAN, bank details — aur usse aapke naam par fraud kiya jata hai.

6.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Deepak ko ek din bank ka message aaya: *“Aapke loan ki Rs. 25,000 ki EMI baki hai.”*

Deepak ne koi loan nahi liya tha! Jaankari lene par pata chala:

1. Kisi ne Deepak ke **Aadhaar aur PAN** ki copy ka istemal karke ek loan online liya tha.
2. Loan ki raqam ek **fake bank account** me gayi.
3. Deepak ka **credit score** kharab ho gaya.

Deepak ne shayad apne documents ki photo kisi fake site par daali thi. **Kisi aur ki personal identity chura kar uske naam par fraud karna — Identity Theft.**

6.2 Definition

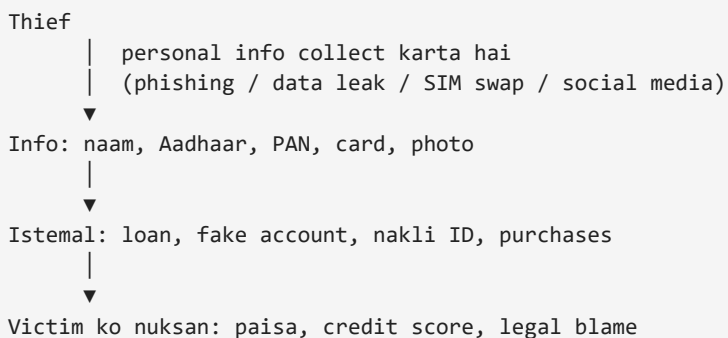
DEFINITION

Identity Theft kisi aur ki **personal identity information** (naam, Aadhaar, PAN, bank/card details, photos, phone number) ko bina permission ke churana aur usse **fraud ya doosre crimes** karne ke liye istemal karna hai.

Simple: Aapka mask pehen kar aapke naam par lootna.

6.3 Flowchart – Identity Theft Process

FLOWCHART



6.4 Identity Info Kaise Chori Hoti Hai (Methods)

- **Phishing / fake sites** — documents maang kar nikalna.
- **Data breaches** — company ka data leak.
- **SIM swap** — SIM block kar ke OTP churana.
- **Skimming** — ATM/card reader se card data.
- **Social media oversharing** — Aadhaar/photos public karna.
- **Dumpster diving** — pheke hue documents se info.

6.5 Impact (Nuksan)

Impact	Kya hota hai
Financial loss	Aapke naam par loan / purchases

Credit score kharab	Future loan mushkil
Legal problems	Crime aapke naam par ho gaya
Reputation	Lokjivan me badnaami

6.6 Prevention (Bachao Ke Tarike)

- Documents (Aadhaar/PAN) ko **public share na karo**.
- **OTP kabhi share na karo**.
- Bank statements / cards ko **regularly check** karo.
- Lost documents ki **turant report** karo.
- Social media par **privacy settings** tight karo.
- Documents ko pheko to **shred** karo.
- **Credit report** time-to-time check karo.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Identity Theft = kisi ki personal identity info chura kar uske naam par fraud.
2. Methods = phishing, data breach, SIM swap, skimming, oversharing.
3. Golden rule: documents + OTP kabhi public/share mat karo.

Viva Q1. Identity theft me kya chori hota hai?

Ans: Personal identity (Aadhaar, PAN, bank info, photos).

Viva Q2. Identity theft hone par victim ko kya nuksan?

Ans: Paisa, credit score, legal blame, reputation.

MEMORY TRICK

Identity Theft = Nakli mask. Jaise koi aapka mask pehen kar aapke naam par lootta hai, waise thief aapki identity pehen kar fraud karta hai.

PART 7 – CYBER TERRORISM

Cyber Terrorism me motive **paisa nahi** hota — motive hota hai **dar, tabahi aur desh ko nuksan**. Isliye ye normal cyber crime se zyada khatarnak hai.

7.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek group kisi desh ki **power company ke control system** par cyber attack karta hai.

Attackers ne:

1. Power grid ke software me virus dal diya.
2. Lakhon gharon ki bijli band kar di.
3. Hospitals, airports aur metro bhi affected hue.

Poori public dar me aa gayi. Government ko emergency declare karni padi.

Ye attack **paisa lene ke liye nahi** tha — motive tha **desh ko dhamkana aur nuksan pahunchana**.

Isi tarah ka aur example: Government ki website par terror ka message likh kar deface karna, ya army ke defense secrets churana. **In sabko Cyber Terrorism kehte hain.**

7.2 Definition

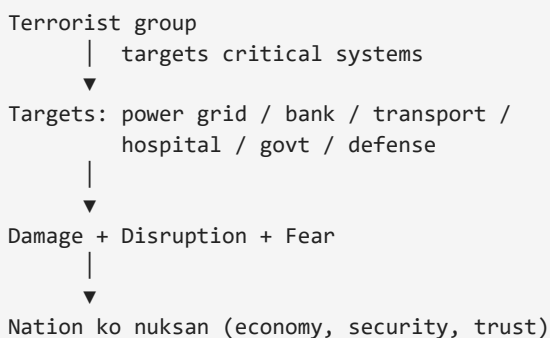
DEFINITION

Cyber Terrorism politically ya ideologically motivated attack hai jo **computer systems, networks ya digital infrastructure** par kiya jata hai — jiska maqsad **dar (fear), bade-scale damage, ya kisi nation / critical infrastructure ko nuksan** pahunchana hai.

Simple: Paisa nahi, **desh aur logon ka dar** lootna.

7.3 Flowchart

FLOWCHART



7.4 Targets of Cyber Terrorism

Target	Kyu
Power grid	Bijli band karne se poori public par asar
Banking network	Economy ka theka system
Transport (airport/rail)	Traffic aur safety
Hospitals	Records aur emergency services
Government websites	Deface + information leak

Defense systems	Secret military data
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7.5 Cyber Crime vs Cyber Terrorism

Basis	Cyber Crime	Cyber Terrorism
Motive	Paisa / personal gain	Fear + political/ideological
Target	Person / company	Nation / critical infrastructure
Effect	Individual loss	Large-scale social/national harm
Who	Cyber criminals, gangs	Terror groups, extremist states

7.6 Prevention (Bachao Ke Tarike)

- **Critical infrastructure protection** — power/bank/defense ki apni security.
- **National cyber security strategy** aur agencies (CERT).
- System par **continuous monitoring** aur threat intelligence.
- **International cooperation** — countries mil kar attack rokna.
- **Redundancy / backups** — ek system fail ho to doosra chal sake.
- Cyber attack hone par **quick response team**.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Cyber Terrorism = dar aur political motive se critical systems par attack.
2. Target = power grid, banks, transport, defense, govt.
3. Cyber Crime = paisa, Cyber Terrorism = nation ko dar/nuksan.

Viva Q1. Cyber terrorism ka motive kya hai?

Ans: Fear + political/ideological motive, nation ko nuksan.

Viva Q2. Cyber crime aur cyber terrorism me farak?

Ans: Crime ka motive paisa, terrorism ka motive dar aur desh ko nuksan.

MEMORY TRICK

Cyber Terrorism = Digital aatank. Jaise aatanki darail ke liye bomb phodta hai, waise cyber terrorist nation ke systems par hamla kar ke dar phailata hai.

PART 8 – VIRTUAL CRIME

Virtual Crime **virtual world** me hone wale crimes hain — online games, metaverse, virtual economies. Isme koi physical cheez nahi hoti, par **real loss** hota hai.

8.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Karan ek online game khelta hai jisme log virtual coins se **virtual items** kharidte hain (sword, skins, pets) jo real paise se kharede jaate hain.

Ek hacker ne Karan ke game account ko hack kiya aur:

1. Uske **virtual coins aur items** chura liye.
2. Unhe real currency me bech diya.
3. Karan ke account se fake chats bhej kar doosre players ko bhi dhoka diya.

Karan ne real paise lagaye the — lekin koi physical cheez chori nahi hui. Fir bhi use **real loss** hua.

Virtual world me hone wala crime (chori, dhoka, harassment) — Virtual Crime.

8.2 Definition

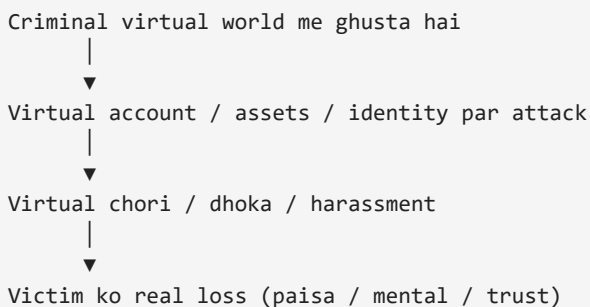
DEFINITION

Virtual Crime wo criminal acts hain jo **virtual environment** (online games, metaverse, virtual worlds, cyberspace) me kiye jaate hain — jisme physical object nahi hota, lekin victim ko **real financial ya emotional loss** hota hai.

Simple: Asli duniya ka dhoka, virtual duniya ka maidan.

8.3 Flowchart

FLOWCHART



8.4 Types of Virtual Crimes

Type	Kya hota hai
Virtual theft	Virtual coins/items/currency churana
Virtual harassment	Avatar/game me dhamkana, gali dena
Virtual fraud	Virtual cheezon ke naam par paisa dhoka
Money laundering	Real paisa virtual currency me white karna
Virtual identity theft	Virtual pehchan/account churana
Virtual property crime	Virtual land/items par kabza

8.5 Important Points

IMPORTANT POINTS

1. Virtual crime me **physical object nahi hota**, par loss real hota hai.
2. **Jurisdiction problem** — virtual world me kiska law lagega, clear nahi.
3. Virtual property ka **ownership** bhi legal issue hai.
4. Isliye game companies account security aur reporting systems rakhti hain.

8.6 Prevention (Bachao Ke Tarike)

- Game / virtual account par **strong password + 2FA**.
- Virtual items real paise se kharidte waqt **trusted platforms** hi use karo.
- Kisi ko bhi **account login share na karo**.
- Suspicious **in-game links** par click na karo.
- Virtual dhande me dhoka ho to **platform ko report** karo.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Virtual Crime = virtual world (games/metaverse) me hone wala crime, real loss ke saath.
2. Types = virtual theft, virtual fraud, harassment, money laundering.
3. Challenge = jurisdiction aur virtual property ka ownership clear nahi.

Viva Q1. Virtual crime me physical object kyu nahi hota?

Ans: Kyunki wo virtual world me hota hai — lekin loss real hota hai.

Viva Q2. Virtual crime ka legal challenge kya?

Ans: Kaunsa desh/law lagega (jurisdiction) aur virtual property ka ownership.

MEMORY TRICK

Virtual Crime = Dream me chori. Jaise sapne me chori ho aur jagne par asar ho — waise virtual world me chori hoti hai aur real loss hota hai.

PART 9 – PERCEPTION OF CYBER CRIMINALS

Perception ka matlab hai — log kisi ko kaise dekhte hain. Ek hi hacker ko ek jagah **hero** mana jata hai, doosri jagah **criminal**. Is part me hum cyber criminals ke types aur unke baare me logon ki soch samjhenge.

9.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek 17 saal ke ladke **Arjun** ne apne school ki website hack kar ke marks change kar diye.

Logon ki alag-alag soch:

- Uske **doston** ko wo **genius / hero** lagta hai — “kaisa cool hack kiya!”
- **School** ke liye wo **criminal** hai — exam system ke saath dhoka.
- **Media** ne usse “super smart hacker” likh diya.
- **Police / cyber cell** usse **juvenile cyber criminal** maanti hai.

Ek hi ladka — **perception** ke hisaab se hero, criminal, ya specialist. Isi tarah **hackers, insurgents aur extremist groups** ko bhi log alag tarah se dekhte hain.

9.2 Definition

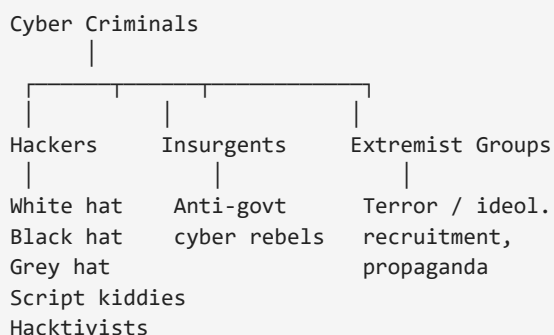
DEFINITION

Perception of Cyber Criminals — society, media, law enforcement aur public ka **nazariya** hai un logon ke baare me jo cyber crimes karte hain — unke **motives, backgrounds, skills aur threat level** ke hisaab se.

Simple: Cyber criminal ko log kaise dekhte hain — hero, criminal, ya threat — motive aur tarah ke hisaab se.

9.3 Flowchart – Cyber Criminals Ki Perception Categories

FLOWCHART



9.4 Hackers (Detail Me)

Type	Kya karta hai	Perception
White Hat	Legal + ethical hacker, security ke liye	Hero / security expert
Black Hat	Criminal motive — data/money chori	Criminal
Grey Hat	Be-ijazat ghusta hai par bina nuksan	Mixed perception
Script Kiddie	Ready tools se khelne wala naya banda	Naive / nuisance
Hactivist	Social/political protest ke liye hack	Kuch ko hero, kuch ko criminal

9.5 Insurgents (Detail Me)

Insurgents wo groups hain jo apni **government ke khilaf** ladte hain — aur cyber ka istemal karte hain:

- Government ki websites aur systems par attack.
- Apna **propaganda** phailana aur logon ko bhadkana.
- Information ko **leak** karke sarkar ko badnaam karna.

Perception: government inhe **criminal/terrorist** maanti hai; inke supporters inhe **freedom fighters** keh sakte hain.

9.6 Extremist Groups (Detail Me)

Extremist groups (terror organizations) internet ka istemal karte hain:

- **Recruitment** — naye logon ko apne me shaamil karna.
- **Propaganda & radicalization** — dushman ke khilaf nafrat phailana.
- **Funding** — donations / crypto se paise jama karna.
- **Attacks** — critical systems par hamla (cyber terrorism).

Perception: pure **terrorist threat** — national security ke liye khatra.

9.7 Baaki Cyber Criminals (Ek Nazar)

- **Organized criminals** — gangs jo financial fraud karti hain.
- **Insiders** — company ke andar ka employee, andar se data leak.
- **Disgruntled employees** — badla lene wale ex-employees.
- **State-sponsored actors** — desh ke support se spying/attacks.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Perception = log cyber criminals ko kaise dekhte hain (hero ya criminal).
2. Hackers: White hat (legal/hero), Black hat (criminal), Grey hat (beech), Script kiddie, Hactivist.
3. Insurgents = anti-government cyber rebels; Extremist groups = terror/ideological (recruitment, propaganda, attacks).

Viva Q1. White hat aur black hat hacker me farak?

Ans: White hat legal/ethical security ke liye; black hat criminal motive se attack karta hai.

Viva Q2. Extremist groups internet kaise use karte hain?

Ans: Recruitment, propaganda, funding, aur cyber attacks.

MEMORY TRICK

Perception = Raai ka nazariya. Jaise ek hi insaan kisi ko hero, kisi ko criminal lagta hai — waise hi hacker bhi motive ke hisaab se alag-alag perception paata hai. Trick: **White=legal, Black=criminal, Grey=beech me.**

PART 10 – WEB SERVER HACKING

Web Server Hacking me attacker **web server par unauthorized access** le leta hai — jis par websites chalti hain. Fir wo data, website ya server ko apne control me le leta hai.

10.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ek company ki website **schoolonline.com** thi jisme students login karte the. Login page par **SQL Injection** ki vulnerability thi.

Hacker ne login box me normal password ki jagah **special SQL code** daal diya:

- Database ne attacker ka code asli command maan liya.
- **Saari students ki table** (usernames, passwords, Aadhaar, parents info) dump ho gayi.
- Hacker ne website ka **home page deface** kar diya.
- Ek **backdoor** bhi install kar diya taaki baar-baar andar aa sake.

Web server (website) par unauthorized access le kar data churana, deface karna ya backdoor daalna — Web Server Hacking.

10.2 Definition

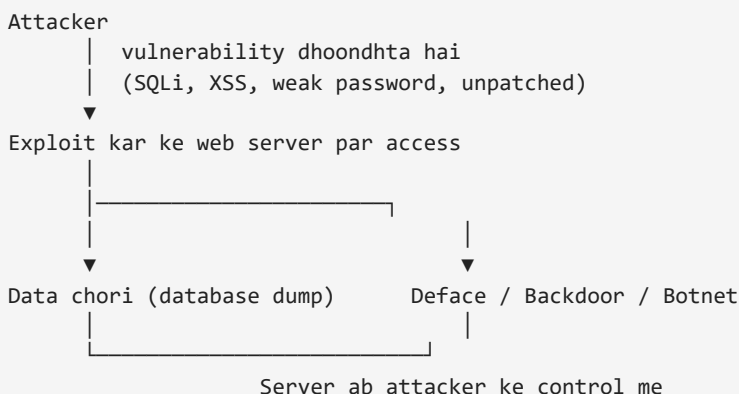
DEFINITION

Web Server Hacking kisi **web server** (jis par websites chalti hain) par **unauthorized access** lena hai — data steal karne, websites deface karne, malware/backdoor install karne, ya server ko aage ke attacks ke liye use karne ke liye.

Simple: Website ki building (server) ka darwaza tod kar andar ghusna.

10.3 Flowchart

FLOWCHART



10.4 Common Methods (Vulnerabilities)

Method	Kya hota hai
SQL Injection	Login/form me SQL code daal kar database se data nikalna
XSS (Cross-site Scripting)	Website par malicious script daal kar users se data lena
Directory Traversal	Server ki secret files tak rasta dhoondhna
Default / weak credentials	admin/admin, 123456 jaisi passwords
Unpatched software	Purane server/plugin me known bug

Misconfiguration

Galat settings se files/backups public ho jana

10.5 Impact (Nuksan)

- **Data breach** — user data market me.
- **Website defacement** — reputation kharab.
- **Backdoor / botnet** — server ab attacker ka.
- **Financial loss** — fines, recovery, business band.
- **Legal action** — data privacy law (DPDP) ke under.

10.6 Prevention (Bachao Ke Tarike)

- **Input validation** — SQL injection/XSS ko roko (parameterized queries).
- Server/software ko **patch & update** karte rehna.
- **Strong admin passwords + 2FA.**
- **WAF** (Web Application Firewall) lagaana.
- **Least privilege** — server files ko kam log hi access karein.
- Website/backups ko **monitor aur test** karte rehna.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Web Server Hacking = web server par unauthorized access le kar data/website/server ko control karna.
2. Sabse aam tarika = SQL injection, XSS, weak password, unpatched software.
3. Prevention = input validation, patching, WAF, strong password.

Viva Q1. SQL injection me kya hota hai?

Ans: Login/form me SQL code daal kar database se data nikalna.

Viva Q2. Defacement kya hai?

Ans: Website ka content bina permission ke badal dena.

MEMORY TRICK

Web Server Hacking = Building ka gate todna. Jaise darwaza tod kar andar ka maal nikalte hain, waise server ka gate (vulnerability) tod kar data nikalte hain.

PART 11 – SESSION HIJACKING

Session Hijacking me attacker **aapki active login session** chura leta hai — password janne ki zaroorat nahi, bina password ke aapke account me ghus jata hai.

11.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Sneha ek cafe ki WiFi par baith kar apne **bank account** me login karti hai.
Login hone par server usse ek **session cookie** deta hai (chhoti file jisse server pehchanta hai ki Sneha logged-in hai).
Usi WiFi par ek attacker **packet sniffer** se wo cookie pakad leta hai — kyunki connection secure nahi tha.
Ab attacker apne browser me wo cookie daal deta hai — aur **bina password ke usi logged-in session par ghus jata hai**.
Wo paise transfer kar deta hai — jab tak Sneha ko pata chalta hai, nuksan ho chuka hota hai.

Kisi user ke active session (cookie/session ID) ko chura kar uske naam par ghusna — Session Hijacking.

11.2 Definition

DEFINITION

Session Hijacking kisi logged-in user ke **active session identifier** (session ID / cookie) ko churana hai — aur usse use kar ke **bina password ke user ke account par take-over** karna hai.

Simple: Aapka ticket (session) chura kar aapki jagah train me baith jaana.

11.3 Flowchart

FLOWCHART

```

graph TD
    A[User login karta hai] --> B[server session ID / cookie deta hai]
    B --> C[Attacker session ID pakadta hai  
(sniffing / XSS / fixation / prediction)]
    C --> D[Attacker session ID ko use karta hai]
    D --> E[Server samajhta hai: yehi user hai]
    E --> F[Attacker user ban jaata hai (take-over)]
  
```

11.4 Types of Session Hijacking

Type	Kya hota hai
Session Sidejacking	Network par packet sniffing se cookie pakadna
Session Fixation	Attacker khud session ID bana kar victim se usse use karwana
Session Prediction	Session ID ka pattern nikal kar agla ID guess karna
Cookie Theft (XSS)	Website par script daal kar cookie nikalwana
Session Replay	Pakda hua request/cookie wapas bhejna

11.5 Prevention (Bachao Ke Tarike)

- **HTTPS** use karo — cookie encrypted tarike se jaaye.

- Cookie par **Secure + HttpOnly** flags — script se nahi pakdi ja sake.
- Login ke baad **session ID regenerate** karo.
- **Session timeout** — kuch der baad session khatam.
- Public WiFi par sensitive kaam na karo.
- **2FA** — session chori hone par bhi extra protection.

EXAM NOTES (2 MARKS / VIVA)**Exam Notes (2 Marks):**

1. Session Hijacking = logged-in user ka session ID/cookie chura kar bina password account take-over.
2. Types = sidejacking, fixation, prediction, cookie theft (XSS).
3. Prevention = HTTPS, Secure+HttpOnly cookie, session regenerate, timeout, 2FA.

Viva Q1. Session hijacking me password ki zaroorat kyu nahi?

Ans: Kyunki attacker server ko wahi session ID de deta hai jo pehle se login hai.

Viva Q2. Session fixation me kya hota hai?

Ans: Attacker apna session ID victim ko use karwa deta hai.

MEMORY TRICK

Session Hijacking = Ticket chori. Jaise koi aapka movie ka ticket chura kar aapki jagah andar ghus jata hai — waise hi session ID chura kar account me ghus jaate hain.

APPENDIX – QUICK REVISION SHEET (UNIT 2)

Sabse Pehle Yaad Rakho — One-Line Definitions

Topic	Ek Line Me
Web Jacking	Website ki poori visitors traffic ko redirect kar ke hamla karna
Online Fraud	Internet par dhokha — card cloning, phishing, advance fee, fake sites
Software Piracy	Software ki illegal copy — activation crack karna, illegal copy banana
Network Intrusion	Network par bina permission ka access — ke andar ghusna
Password Sniffing	Network par password chup-chaap pakadna
Identity Theft	Doosre ki pehchan chura kar uske naam par kaam karna
Cyber Terrorism	Dar + political motive se critical systems par attack
Virtual Crime	Virtual world (games/metaverse) me crime, real loss ke saath
Web Server Hacking	Web server par unauthorized access le kar data/deface/backdoor
Session Hijacking	Logged-in user ka session ID chura kar account take-over

Memory Tricks

- Web Jacking = **Uturna (redirect)** — traffic udta hai.
- Online Fraud = **Digital Dhoka**.
- Software Piracy = **Soft COPY bina license**.
- Network Intrusion = **Andar ghusna**.
- Password Sniffing = **Sungh kar sungha lena**.
- Identity Theft = **Doosre ka mask pehenna**.
- Cyber Terrorism = **Digital aatank**.
- Virtual Crime = **Dream me chori**.
- Web Server Hacking = **Building ka gate todna**.
- Session Hijacking = **Ticket chori**.

Rapid Fire Viva (15 Questions)

Q	Answer
Web jacking me kya hack hota hai?	Website ki traffic / domain
Web jacking ke 3 tarike?	Vulnerability, spam email, hijacked website
Online fraud ke 2 examples?	Card cloning, phishing
Card cloning me kya banaya jata hai?	Duplicate card (skimmed info se)
Phishing me attacker kya banata hai?	Fake page / fake link
Software piracy ke 3 types?	Soft-lifting, counterfeiting, internet piracy
Network intrusion ke 2 types?	External + insider (internal) attack
Intrusion prevention system?	IPS

Password sniffing me attacker kya use karta hai?	Sniffer / packet capture tool
Password sniffing ka desi solution?	HTTPS, 2FA, password manager
Identity theft ke 2 types?	Financial + Criminal identity theft
Identity theft ka sabse bada nuksan?	Aapke naam par illegal kaam + CIBIL kharab
Cyber terrorism ka motive?	Fear + political/ideological motive
Web server hacking ka sabse common method?	SQL injection
Session hijacking me password ki zaroorat?	Nahi — session ID hi kaafi hai

Note: Ye Unit 2 ka complete notes hai. Unit 1 aur 2 ke saare important points isi series me cover ho chuke hain.